Domus Horizon: Discrete Sweep Reports and Audits for the Millennium Problems

Domus Horizon Project — Computational Research Core

July 28, 2026

Abstract

This document consolidates the data trains, metrics, and experimental numerical validations corresponding to the seven Millennium Problems and auxiliary conjectures, processed under the reproducible discrete execution engine (v003). The presented results constitute discrete and computationally bounded explorations, operating as high-density simulations and approximations for peer audit, without constituting formal analytical proofs of universal scope.

1 Introduction and Methodological Framework

The set of experimental trials was executed under a controlled environment with an explicit global seed and a deterministic binary mode. Measurements cover discrete scales parameterized up to $N = 1,000,000$ depending on the domain of each problem.

2 Results by Millennium Problem

Table 1: Summary of Millennium Problem Evaluations

Problem	**Domain / Scale**	**Key Metric**	**Methodolo**
A1: P vs NP	VRP Instances ($N = 50,000$)	$\alpha = 0.6996$	Finite empir
A2: Riemann Hypothesis	Truncated zeta sum ($N = 1,000,000$)	Graph spectrum	Discrete app
A3: Hodge Conjecture	KNN-12 Graphs ($N = 5,000$)	$29,562$ H_1 cycles	Bounded top
A4: Yang-Mills	$U(1)$ Annular lattice ($N = 50,000$)	E_0, E_1 gaps	Lattice simu
A5: Navier-Stokes	Cubic Voronoi ($N = 1,000,000$)	No blowup events	Discrete scal
A6: Birch and Swinnerton-Dyer	Elliptic curves ($p < 1000$)	Euler product	Low-rank au
A7: Poincaré Conjecture	S^3, T^3, RP^3 proxies ($N = 5,000$)	Spectral prostheses	Manifold an

2.1 Detailed Experimental Evaluations

- **A1 (P vs NP):** Evaluated on Vehicle Routing Problem (VRP) instances up to $N = 50,000$ with a scaling parameter $\alpha = 0.6996$. *Review note:* Does not prove the equality of complexity classes in a general sense.
- **A2 (Riemann):** Spectral analysis via truncated zeta function sums up to 10^6 terms. *Review note:* Does not validate absolute alignment on the critical line for infinite zeros.
- **A3 (Hodge):** Harmonic cycle mapping in nearest-neighbor graphs (KNN-12) identifying homology classes in H_1.

- **A4 (Yang-Mills):** Measurement of the spectral mass gap (E_0 to E_1) in $U(1)$ discrete gauge manifolds.
- **A5 (Navier-Stokes):** Numerical integration on a periodic three-dimensional Voronoi mesh, evaluating the boundedness of enstrophy and absence of finite-time singularities.
- **A6 (BSD):** Verification of Euler product behavior on elliptic curves with reduced conductor ($11a_1$, $14a_1$, $15a_1$, etc.).
- **A7 (Poincaré):** Characterization of the discretized Laplace-Beltrami operator over reference topological spaces of dimension 3.

3 Complementary Conjectures and Number Theory

In addition to the seven core problems, verification records for fundamental arithmetic structures were included:

Table 2: Auxiliary Conjecture Metrics

Conjecture	**Parameter / Limit**	**Observed Result**
B: Ulam Spiral	$46,012,125$ integers	$5,924$ on diagonals ($A_{\text{obs}} = 2.0770$)
C1: Goldbach	$N = 100,000$	$49,999$ verified pairs (no counterexamples)
C2: Twin Primes	Limit N	$\pi_2(N) = 1,224$ (partial Brun 1.6728)
C3: Collatz	$N = 1,000,000$	Max 524 steps at $n = 837,799$

4 Conclusions

The data trains from run `0d17ab72b786` confirm the computational stability and internal consistency of the evaluated discrete models. The methodological framework is documented for replication and peer review in high-performance computing environments.